MAIS PAC

IPhO 04.01: Algebra, Functions and Geometry · Worked Solutions

Algebra, Functions and Geometry: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IPhO questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

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Solution 1 — Completing the square. Return →

(a) R is largest when sin 2θ = 1, that is 2θ = 90°, so θ = 45°.

(b) sin 2θ = (Rg)/(v²) = (40×9.81)/(400) = 0.981, so 2θ = 78.8° or 101.2°, giving θ = 39.4° or 50.6°. The two angles are complementary, as they must be, because sin 2θ is symmetric about 45°. **Check:** the maximum range at this speed is v²/g = 40.8 {m}, so 40 {m} is just inside the reachable range — which is why the two angles are close to 45°.

Solution 2 — A ratio from similarity. Return →

(a) (H)/(12.0) = (1.80)/(2.40), so H = 9.0 {m}.

(b) Both are right triangles, and the Sun's rays are parallel, so the angle of elevation is the same at both sites. Two equal angles are enough for similarity.

(c) The ground may not be level, the pole may not be vertical, and the shadows may not be measured at the same instant, so the Sun angle may differ slightly between the two sites. **Check:** a difference of 1° in the Sun's altitude changes the ratio by about 2%, which is 0.2 {m} on a 9 {m} tree.

Solution 3 — Two equations, two unknowns. Return →

(a) With a the acceleration of m_2 downwards: m_2g - T = m_2a and T - m_1g = m_1a.

(b) Adding eliminates T: (m_2-m_1)g = (m_1+m_2)a, so a = (2.0)/(8.0)×9.81 = 2.45 {m,s^(-2)}. Then T = m_1(g+a) = 3.0×12.26 = 36.8 {N}.

(c) With m_1 = m_2 = m, a = 0 and T = mg: the string simply supports each mass. **Check:** when m_1→0, a→ g and T→0, as a free-falling mass requires.

Solution 4 — The cosine rule in a field problem. Return →

(a) Each force points along a side away from a vertex of the equilateral triangle, so the angle between the two force vectors is 60°.

(b) Each force has magnitude F = kq²/d². Adding two equal vectors at 60°: R = 2Fcos 30° = √(3),kq²/d².

(c) Bringing the fixed charges closer shrinks d, so each force grows as 1/d² **and** the geometry stays equilateral, giving R ∝ 1/d² for fixed q — but if the experiment keeps the total charge fixed while d changes, the potential energy ∝ 1/d grows faster in relative terms. **Check:** the factor √(3) is the classic triangular-lattice result and reappears in the geometry of close-packed crystals.

Solution 5 — Small-angle approximations. Return →

(a) The restoring force is mgsin θ along the arc, and the displacement is s = Lθ. For small θ, sin θ≈θ, so F ≈ -(mg/L)s: a linear restoring force, hence simple harmonic motion with ω = √(g/L).

(b) θ = 5.0° = 0.0873 {rad}, and sin θ = 0.08716. The fractional difference is (0.08727-0.08716)/0.08727 = 1.3×10^(-3), about 0.13%.

(c) With sin θ ≈ θ - θ³/6, the error is θ²/6 = 0.01, so θ = 0.245 {rad} = 14°. **Check:** the period error goes as θ²/16, a quarter of this, so a 5° pendulum keeps time to about 0.03% — far better than a human reaction time.

Solution 6 — A discriminant condition. Return →

(a) With $y = h + v\sin\theta\, t - \tfrac{1}{2}gt^2$ and $y = 0$: $\tfrac{1}{2}gt^2 - v\sin\theta\, t - h = 0$.

(b) The discriminant is $v^2\sin^2\theta + 2gh$, which is a sum of two non-negative terms and is therefore positive for any $h>0$ or any non-zero launch speed. So there are two real roots.

(c) Physically, a projectile launched from a height always reaches the ground, whatever the angle — the two roots are the "now" root and the "later" fall. A negative discriminant would mean the projectile never arrives, which happens only if the ground is replaced by a curved Earth and the projectile enters orbit, or if you try to solve for a time before launch. **Check:** one root is negative, corresponding to the fictional past trajectory that would have passed through the launch point from below.

Solution 7 — Inverse-square geometry. Return →

(a) $I \propto 1/r^2$, so at $2r$ the intensity is $1/2^2 = 1/4$.

(b) By symmetry the point is midway: 2.0 {m} from each lamp. Because the lamps are identical, the condition $P/(4\pi x^2) = P/(4\pi(4-x)^2)$ reduces to $x = 4-x$.

(c) The power cancels from the equation: the equality of illuminance depends only on the two distances. If the lamps differ, the point moves towards the weaker one, and the balance condition becomes $P_1/x^2 = P_2/(d-x)^2$. **Check:** with $P_1 = 4P_2$ the balance point is at $x = 2d/3$ from the stronger lamp, which is the working principle of a photometric bench.

Solution 8 — Minimising a function. Return →

(a) Resolving vertically: $N = mg - F\sin\theta$. Constant speed needs $F\cos\theta = \mu N = \mu(mg - F\sin\theta)$, so

$$F = \frac{\mu\, mg}{\cos\theta + \mu\sin\theta}. \tag{1}$$

(b) F is smallest when the denominator is largest. Setting the derivative of $\cos\theta+\mu\sin\theta$ to zero gives $-\sin\theta+\mu\cos\theta = 0$, that is $\tan\theta = \mu$. (Alternatively write the denominator as $\sqrt{1+\mu^2}\cos(\theta-\varphi)$ with $\tan\varphi = \mu$, maximised at $\theta = \varphi$.)

(c) With $\mu = 0.40$, $\theta = 21.8^\circ$ and $\cos\theta+\mu\sin\theta = 0.928+0.149 = 1.077$, so $F = 0.40\times10\times9.81/1.077 = 36$ {N}. Pulling horizontally would need $0.40\times98.1 = 39$ {N}. **Check:** pulling upwards at the friction angle lightens the normal force exactly as much as it loses in horizontal component — the classic optimum.

Solution 9 — Three equations, one result. Return →

(a) Let N_w be the wall force (horizontal), N_f the floor normal force and f the friction. Then N_w = f, N_f = Mg, and taking moments about the floor end: N_wLsin θ = Mg(L)/(2)cos θ.

(b) From the moment equation N_w = (Mg)/(2)θ, so f = (Mg)/(2)θ. The friction law requires f μ Mg, hence (1)/(2)θ μ.

(c) As θ→90° (rod nearly vertical) θ→0 and almost no friction is needed. As θ→0 the required friction diverges, so a rod cannot rest at a very shallow angle on any floor with finite μ. **Check:** at μ = 0.5 the critical angle is θ = 45°, a sensible number for a wooden rod on a wooden floor.

Solution 10 — An identity in disguise. Return →

(a) Asin (ω t) + Asin (ω t+φ) = 2Acos ()/(){φ}{2}sin (ω t + (φ)/(2)).

(b) The amplitude is 2Acos ()/(){φ}{2}.

(c) The resultant vanishes when cos ()/(){φ}{2} = 0, that is φ = π (or 180°): the waves are perfectly out of phase and cancel everywhere. **Check:** the result is the simplest case of interference, and it shows that the combined amplitude can be anywhere from 0 to 2A — energy is redistributed, not lost, since the intensity $4A^2$cos ^2(φ/2) peaks at φ = 0.